



SIMATS
ENGINEERING



SIMATS
Saveetha Institute of Medical And Technical Sciences
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

Application Based Questions

Rakshitha V B

192324004

B.Tech AI&DS

1. Student Admission Data Cleaning

A university has received a CSV file containing student admission records collected from multiple departments. The file contains duplicate records, missing phone numbers, inconsistent capitalization of department names, and extra spaces in student names. Write a Python program to read the CSV file, perform appropriate data wrangling operations to clean and standardize the data, remove duplicate records, and save the cleaned dataset into a new CSV file suitable for further analysis.

Code:

```
import pandas as pd

df = pd.read_csv("student_admission.csv")

df["Student_Name"] = df["Student_Name"].str.strip().str.title()

df["Department"] = df["Department"].str.strip().str.title()

df["Phone_Number"] = df["Phone_Number"].fillna("Not Available")

df = df.drop_duplicates()

df.to_csv("student_admission_cleaned.csv", index=False)

print("Original Dataset")

print(pd.read_csv("student_admission.csv"))

print("\nCleaned Dataset")

print(df)
```

Output:

```
6      106      Sneha Reddy      Mechanical  9.871235e+09
7      102      Anita Verma      information Technology      NaN
8      107      Ravi Kumar      ELECTRONICS AND COMMUNICATION  9.876541e+09
```

Cleaned Dataset

	Student_ID	Student_Name	Department	Phone_Number
0	101	Rahul Sharma	Computer Science	9876543210.0
1	102	Anita Verma	Information Technology	Not Available
2	103	Karan Singh	Mechanical	9123456789.0
4	104	Priya Nair	Electronics And Communication	Not Available
5	105	Arjun Patel	Computer Science	9988776655.0
6	106	Sneha Reddy	Mechanical	9871234567.0
8	107	Ravi Kumar	Electronics And Communication	9876541111.0

2. Employee Data Integration

A company maintains employee details in a CSV file and department information in a JSON file. Both files contain a common Department ID field. Write a Python program to read the two files, combine the employee and department information, identify records with missing department details, and generate a consolidated dataset that can be stored in a relational database.

Code:

```
import pandas as pd

employees = pd.read_csv("employees.csv")

departments = pd.read_json("departments.json")

merged = pd.merge(employees, departments, on="Department_ID", how="left")

missing = merged[merged["Department_Name"].isnull()]

print("Employee Dataset")
print(employees)

print("\nDepartment Dataset")
print(departments)

print("\nConsolidated Dataset")
print(merged)
```

```
print("\nEmployees with Missing Department Details")
```

```
print(missing)
```

```
merged.to_csv("employee_department_consolidated.csv", index=False)
```

Output:

```
Employee_ID Employee_Name Department_ID Salary
0          101         Rahul            1   50000
1          102         Anita            2   55000
2          103         Karan            3   60000
3          104         Priya            4   52000
4          105         Arjun            2   58000
5          106         Sneha            5   62000
```

Department Dataset

```
Department_ID Department_Name Location
0             1   Human Resources  Chennai
1             2 Information Technology Bangalore
2             3             Finance  Hyderabad
```

Consolidated Dataset

```
Employee_ID Employee_Name Department_ID Salary Department_Name Location
0          101         Rahul            1   50000   Human Resources  Chennai
1          102         Anita            2   55000 Information Technology Bangalore
2          103         Karan            3   60000             Finance  Hyderabad
3          104         Priya            4   52000                NaN        NaN
4          105         Arjun            2   58000 Information Technology Bangalore
5          106         Sneha            5   62000                NaN        NaN
```

Employees with Missing Department Details

```
Employee_ID Employee_Name Department_ID Salary Department_Name Location
3          104         Priya            4   52000                NaN        NaN
5          106         Sneha            5   62000                NaN        NaN
```

3. Hospital Patient Data Validation

A hospital stores patient information in a CSV file containing patient IDs, names, ages, blood groups, and contact numbers. Some records contain missing values, invalid ages, and duplicate patient IDs. Write a Python program to read the dataset, identify and correct data quality issues, remove duplicate records, and generate a validated dataset that can be imported into the hospital database.

Code:

```
import pandas as pd

df = pd.read_csv("patients.csv")

df["Age"] = pd.to_numeric(df["Age"], errors="coerce")

df.loc[(df["Age"] < 0) | (df["Age"] > 120), "Age"] = None

mean_age = int(df["Age"].mean())

df["Age"] = df["Age"].fillna(mean_age)

df = df.drop_duplicates(subset="Patient_ID")

df.to_csv("validated_patients.csv", index=False)

print("Original Dataset")
print(pd.read_csv("patients.csv"))
```

```
print("\nValidated Dataset")
```

```
print(df)
```

Output:

Original Dataset

	Patient_ID	Patient_Name	Age	Blood_Group	Contact_Number
0	101	Rahul Sharma	25.0	A+	9876543210
1	102	Anita Verma	NaN	-	9876501234
2	103	Karan Singh	-5.0	B+	9988776655
3	104	Priya Nair	130.0	O+	9123456789
4	105	Arjun Patel	45.0	AB+	9871234567
5	101	Rahul Sharma	25.0	A+	9876543210
6	106	Sneha Reddy	32.0	O-	9988112233
7	107	Ravi Kumar	NaN	B-	9876541111
8	108	Meena Das	29.0	A-	9988001122

Validated Dataset

	Patient_ID	Patient_Name	Age	Blood_Group	Contact_Number
0	101	Rahul Sharma	25.0	A+	9876543210
1	102	Anita Verma	31.0	-	9876501234
2	103	Karan Singh	31.0	B+	9988776655
3	104	Priya Nair	31.0	O+	9123456789
4	105	Arjun Patel	45.0	AB+	9871234567
6	106	Sneha Reddy	32.0	O-	9988112233
7	107	Ravi Kumar	31.0	B-	9876541111
8	108	Meena Das	29.0	A-	9988001122